

Existence of Solutions Algebraic Equations from Existence of Solutions of Ordinary Differential Equations



大類昌俊:『数理解析学概論』序文に2回載った人

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Existence theorem of solutions of ordinary differential equations (Cauchy-Lipschitz theorem) follows without existence of algebraic equations. We can proof Existence of solutions algebraic equations from existence of solutions of ordinary differential equations.

Linear ordinary differential eqations with constant coefficients on bounded and closed interval I , $P(D)u(x) = 0, x \in I, P(0) \neq 0$ has an unique solution $u \in C^1(\bar{I})$ from corollary of Cauchy-Lipschitz theorem ([1]). u has an extension $\bar{u} \in \mathcal{S}$. By Fourier transform, we obtain $P(\xi)\mathcal{F}\bar{u}(\xi) = 0 . P(\xi)\mathcal{F}\bar{u}(\xi) = 0 \iff P(D)\bar{u}(x) = 0$ ([2]). $P(0) \neq 0$, therefore $\exists \xi, \mathcal{F}\bar{u}(\xi) \neq 0$, so we have an existence of ξ such that $P(\xi) = 0$.

References: [1]石原茂-矢野健太郎, 解析学概論, 裳華房, 2020; [2]垣田高雄-柴田良弘, ベクトル解析から流体へ, 日本評論社, 2024.